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In [ ]: PAWAR SAKSHI MAHENDRA  
        ROLL NO - 49  
        PRACTICAL NO - 09
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In [1]: import numpy as np  
  
        class HopfieldNetwork:  
  
            def __init__(self, n_neurons):  
                self.n_neurons = n_neurons  
                self.weights = np.zeros((n_neurons, n_neurons))  
  
            def train(self, patterns):  
                for pattern in patterns:  
                    self.weights += np.outer(pattern, pattern)  
  
                # No self-connections  
                np.fill_diagonal(self.weights, 0)  
  
            def predict(self, pattern):  
                # Update state (synchronously)  
                net_input = np.dot(pattern, self.weights)  
                return np.sign(net_input)  
  
        # Main execution  
        if __name__ == '__main__':  
  
            patterns = np.array([  
                [1, 1, -1, -1],  
                [-1, -1, 1, 1],  
                [1, -1, 1, -1],  
                [-1, 1, -1, 1]  
            ])  
  
            n_neurons = patterns.shape[1]  
  
            network = HopfieldNetwork(n_neurons)  
            network.train(patterns)  
  
            for pattern in patterns:  
                prediction = network.predict(pattern)  
                print("Input pattern:", pattern)  
                print("Predicted pattern:", prediction)  
                print()
```

Input pattern: [1 1 -1 -1]
Predicted pattern: [1. 1. -1. -1.]

Input pattern: [-1 -1 1 1]
Predicted pattern: [-1. -1. 1. 1.]

Input pattern: [1 -1 1 -1]
Predicted pattern: [1. -1. 1. -1.]

Input pattern: [-1 1 -1 1]
Predicted pattern: [-1. 1. -1. 1.]

In []: